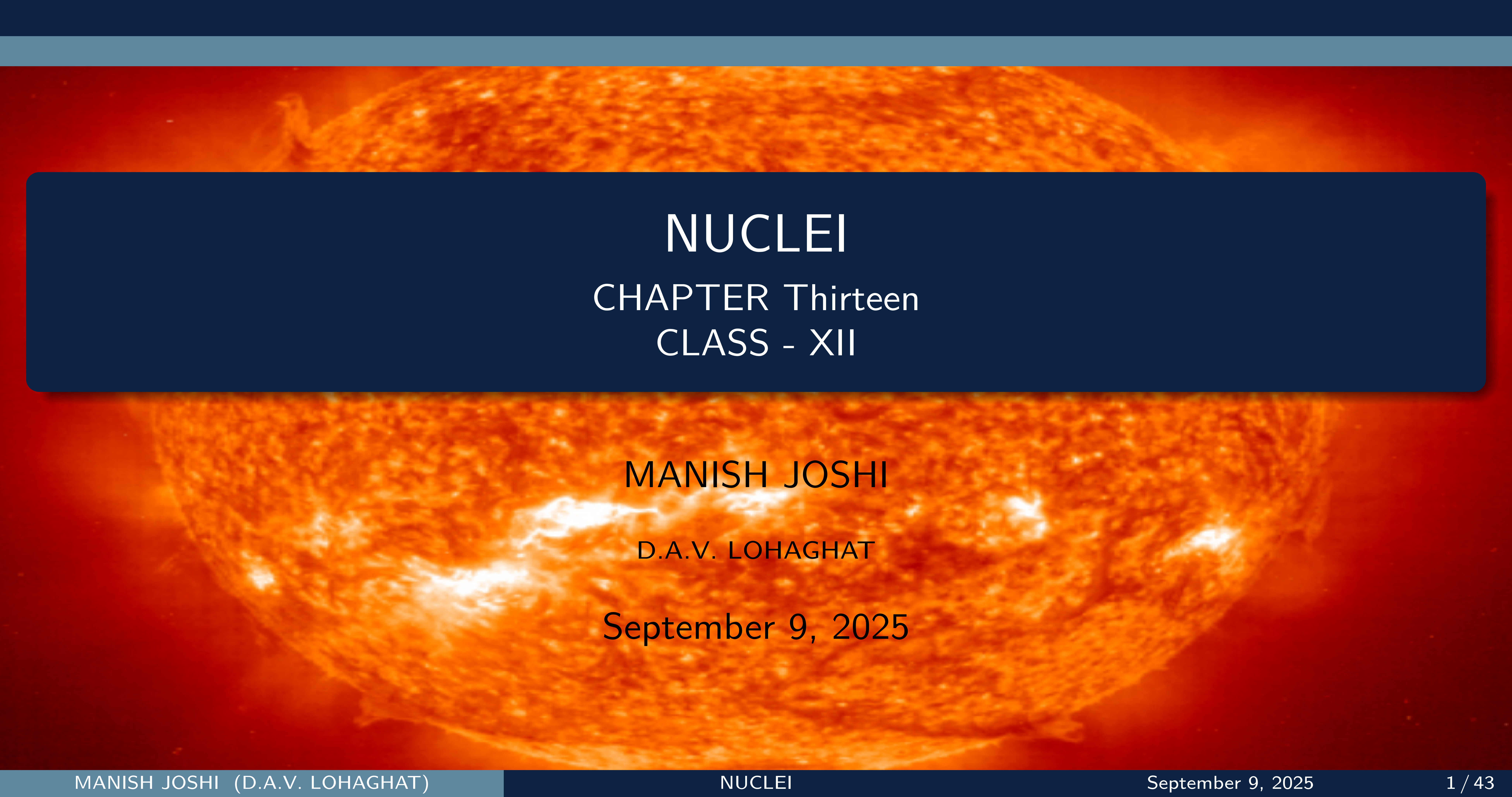




NUCLEI

CHAPTER Thirteen

CLASS - XII

MANISH JOSHI

D.A.V. LOHAGHAT

September 9, 2025

Learning Objectives:

- Composition and size of nucleus, nuclear force
- Mass-energy relation, mass defect; binding energy per nucleon and its variation with mass number;
- Nuclear Fission, Nuclear Fusion.

Introduction

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- This means the **volume of a nucleus** is about **10^{-12}** times the **volume of the atom**. In other words, an **atom is almost empty**.
- Nevertheless, the **nucleus contains most** (more than **99.9%**) of the **mass of an atom**.

ATOMIC MASSES AND COMPOSITION OF NUCLEUS

Atomic masses:

- The **mass of an atom is very small**, compared to a kilogram; for example, the **mass of a carbon atom, ^{12}C , is 1.992647×10^{-26} kg.**

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- **Accurate measurement of atomic masses** is carried out with a **mass spectrometer.**

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- since the atom is neutral total **charge of the atomic electrons** is thus **(-Ze)**

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- Since the **nuclei of deuterium and tritium are isotopes of hydrogen**, they must **contain only one proton** each.

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- This hypothesis was verified in 1932 by **James Chadwick** who **observed emission of neutral radiation** when **beryllium nuclei were bombarded with alpha-particles**

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Discovery of Neutron:

- Chadwick assume that the neutral radiation consists of a new type of neutral particles called **neutrons**. From conservation of energy and momentum, he was able to determine the **mass of new particle as very nearly the same as mass of proton**
- mass of neutron $m_n = 1.00866u = 1.6749 \times 10^{-27} \text{ kg}$

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- Chemical properties of isotopes are identical (same electronic configuration).

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- For example, the nucleus of gold is denoted by ${}^{197}_{79}Au$. It contains 197 nucleons, of which 79 are protons and the rest 118 are neutrons.

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ISOBARS AND ISOTONES

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- **Isotones:** Same neutron number (N), different Z . Example: ${}^{198}_{80}\text{Hg}$ and ${}^{197}_{79}\text{Au}$.

SIZE OF THE NUCLEUS

- Rutherford's suggestion, Geiger and Marsden performed their classic experiment: on the scattering of α -particles from thin gold foils. Their experiments revealed that the distance of closest approach to a gold nucleus of an α -particle of kinetic energy 5.5 MeV is about 4.0×10^{-14} m.

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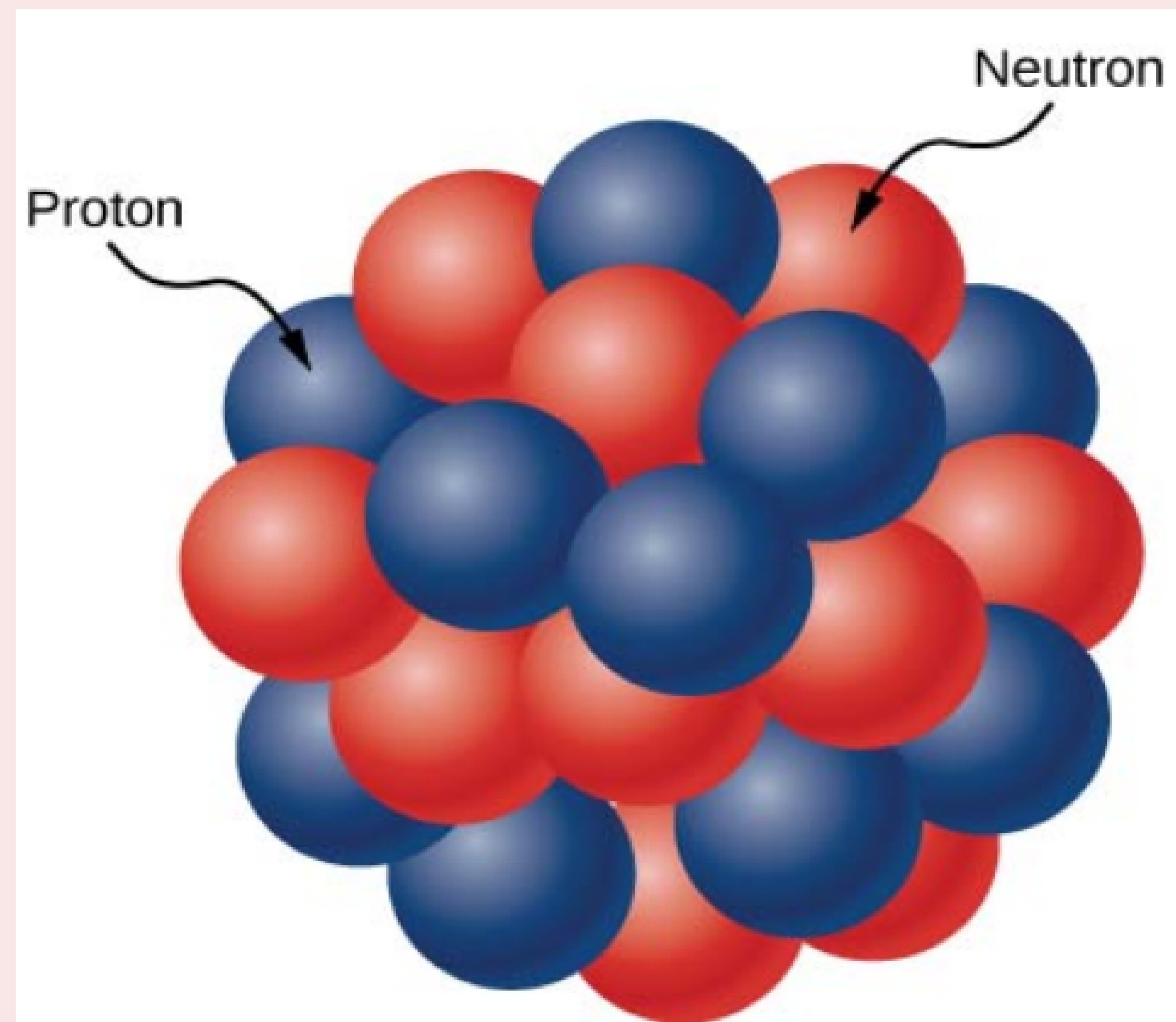
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- the **actual size of the nucleus has to be less than 4.0×10^{-14} m.**
- By performing scattering experiments in which **fast electrons**, instead of α -particles, are projectiles that bombard targets made up of various elements, the **sizes of nuclei of various elements have been accurately measured.**

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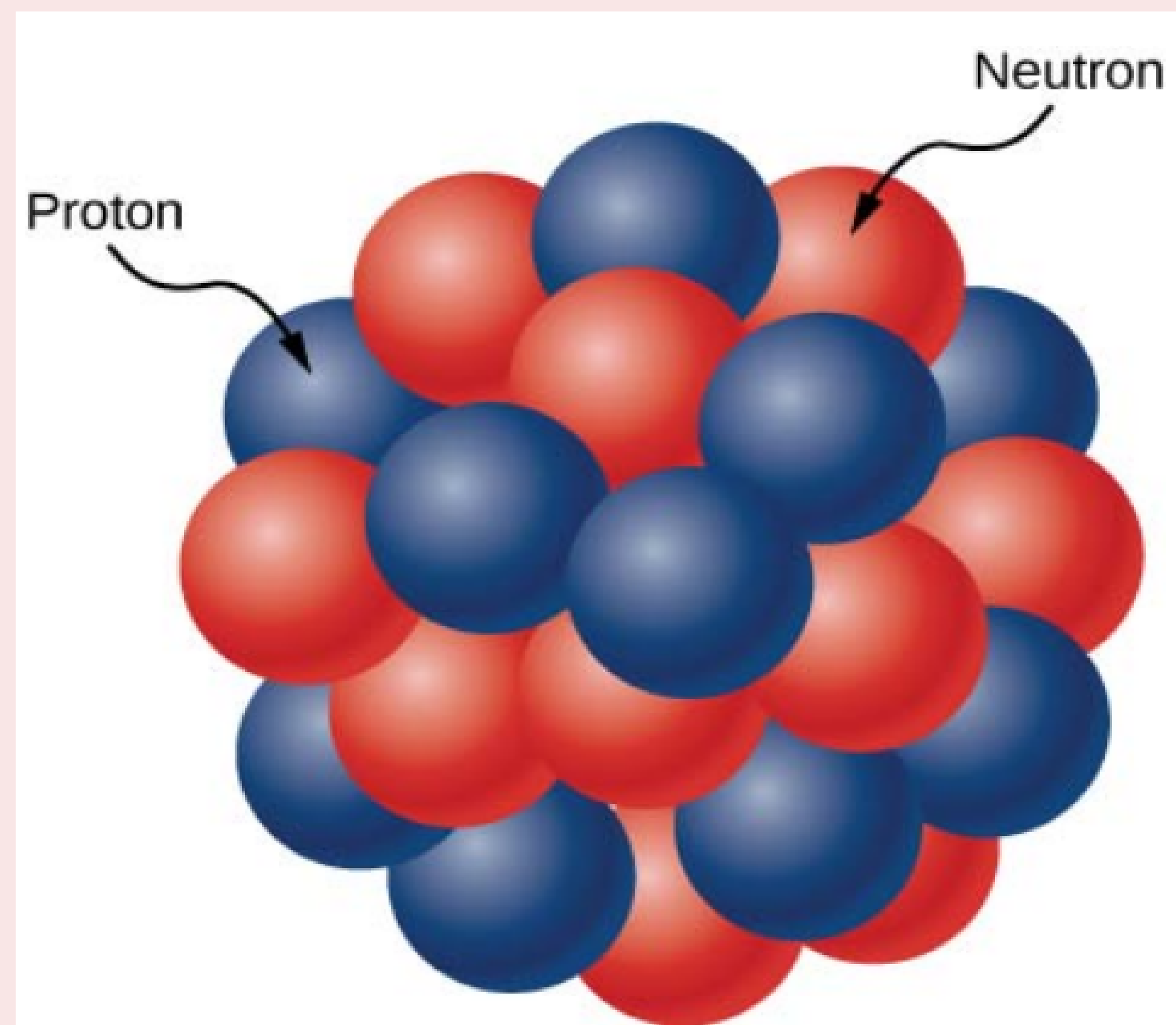
- The simplest model of the nucleus is a densely packed sphere of nucleons.



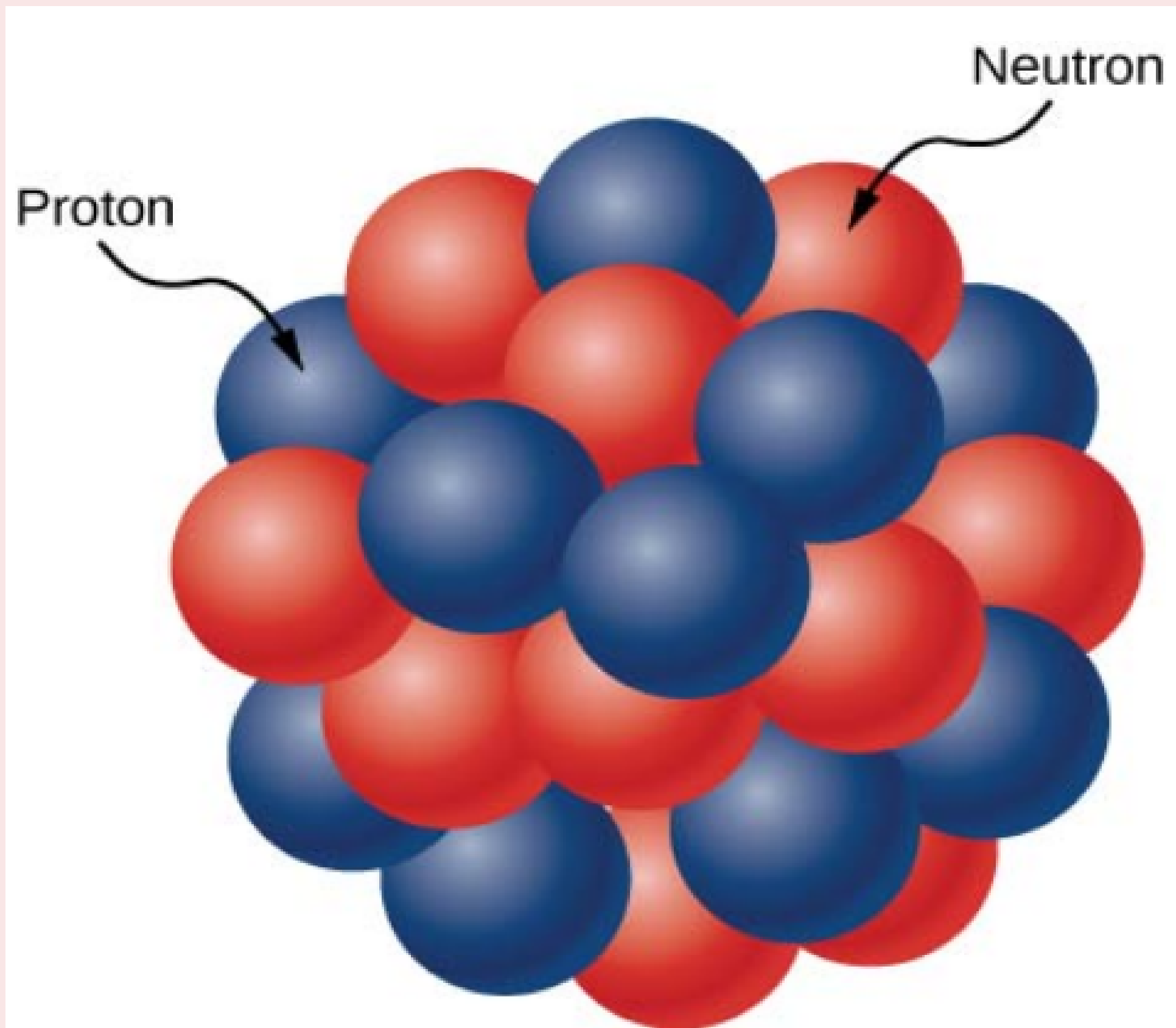
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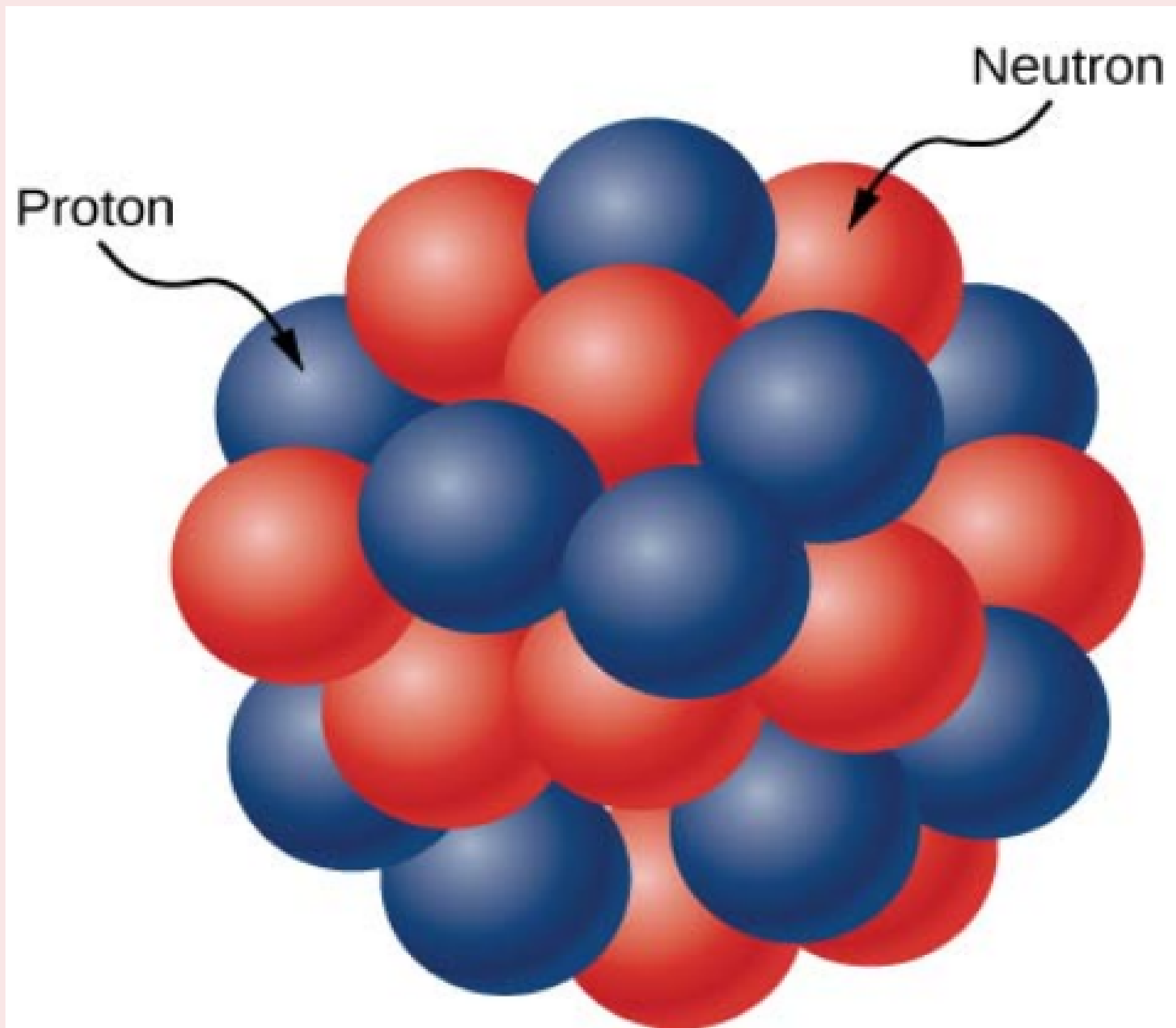


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$$R = R_0 A^{1/3}$$

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- **Density** = $\frac{\text{Mass of nucleus}}{\text{Volume of nucleus}} = \frac{A}{\frac{4}{3}\pi R^3} = \frac{A}{\frac{4}{3}\pi R_0^3 A}$

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- $\text{Density} = \frac{1}{\frac{4}{3}\pi R_0^3} = \text{Constant} = 2.3 \times 10^{17} \text{ kgm}^{-3}$

Example 1

Given the **mass** of iron nucleus as **55.85 u** and **A=56**, find the **nuclear density**?
($1u = 9.27 \times 10^{-26} \text{ kg}$)

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- However, **Einstein** showed that **mass is another form of energy** and **one can convert mass-energy into other forms of energy**, say kinetic energy and vice-versa.
- Einstein gave the famous **mass-energy equivalence relation**

$$E = mc^2$$

Here, m is energy equivalent of mass and c is speed of light in vacuum.

Example 2

Calculate the energy equivalent of 1 g of substance.

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- However, According to nuclear particle experiments, the **total mass of a nucleus (M_{nuc}) is less than the sum of the masses of its constituent nucleons (protons and neutrons).**

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- Thus, we find that the **mass of the ${}^{16}_8\text{O}$ nucleus is less than the total mass of its constituents by 0.13691 u**.

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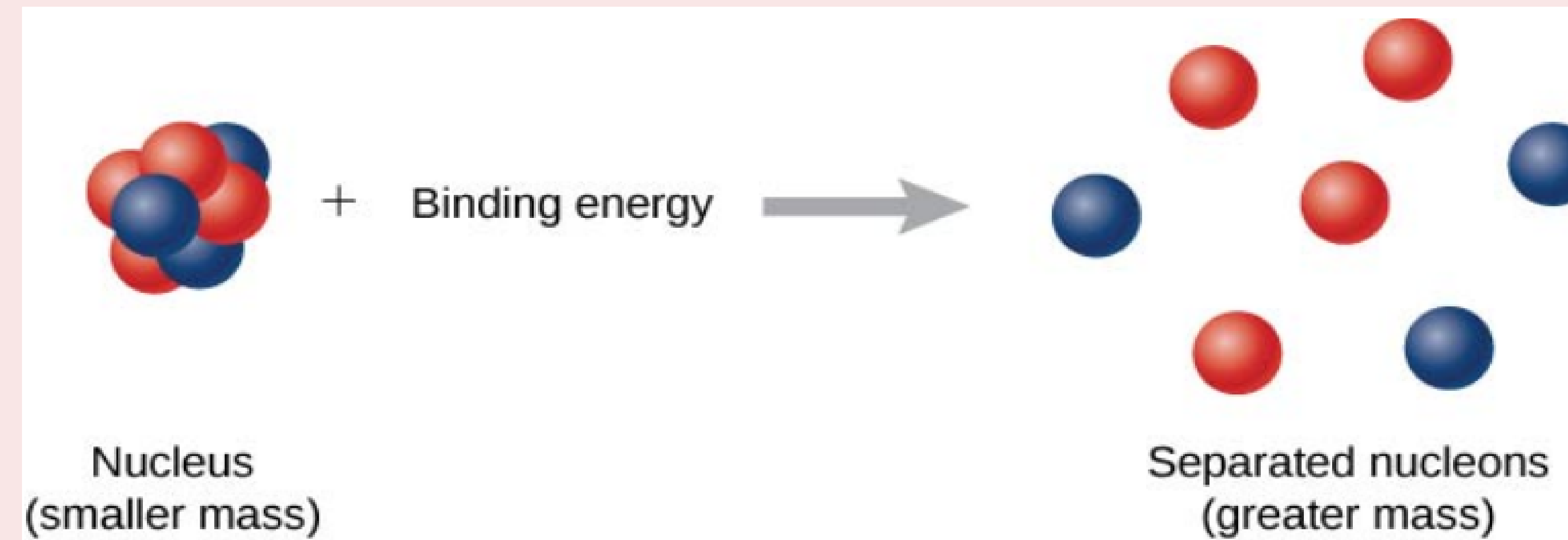
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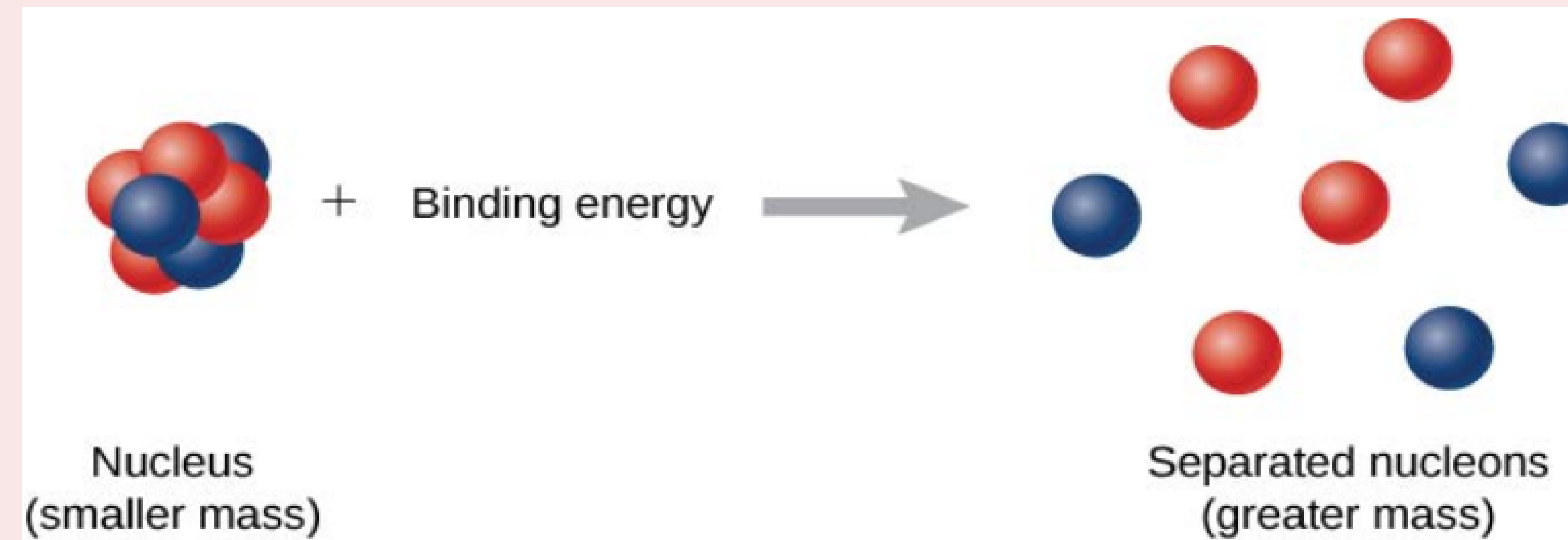
- where $Z M_p$ is the **total mass of the protons**, $(A - Z) M_n$ is the **total mass of the neutrons**, and M_{nuc} is the **mass of the nucleus**.

Binding Energy



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- Now **imagine this process occurs in reverse**. Instead of forming a nucleus, **energy is put into the system to break apart the nucleus**. The amount of energy required is called the **total binding energy (BE)**, ($E_b = \Delta Mc^2$).

Example 13.3 Find the energy equivalent of one atomic mass unit, first in Joules and then in MeV. Using this, express the mass defect of $^{16}_8\text{O}$ in MeV/c^2 .

Solution

$$1\text{u} = 1.6605 \times 10^{-27} \text{ kg}$$

To convert it into energy units, we multiply it by c^2 and find that

$$\text{energy equivalent} = 1.6605 \times 10^{-27} \times (2.9979 \times 10^8)^2 \text{ kg m}^2/\text{s}^2$$

$$= 1.4924 \times 10^{-10} \text{ J}$$

$$= \frac{1.4924 \times 10^{-10}}{1.602 \times 10^{-19}} \text{ eV}$$

$$= 0.9315 \times 10^9 \text{ eV}$$

$$= 931.5 \text{ MeV}$$

$$\text{or, } 1\text{u} = 931.5 \text{ MeV}/c^2$$

$$\text{For } ^{16}_8\text{O}, \quad \Delta M = 0.13691 \text{ u} = 0.13691 \times 931.5 \text{ MeV}/c^2$$

$$= 127.5 \text{ MeV}/c^2$$

The energy needed to separate $^{16}_8\text{O}$ into its constituents is thus $127.5 \text{ MeV}/c^2$.

Graph of Binding Energy per Nucleon

- In nuclear physics, one of the most important experimental quantities is the binding energy per nucleon (E_{bn}), which is the **ratio of the binding energy** E_b of a nucleus **to the number of the nucleons**, A , in that nucleus:

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- We can think of **Binding energy per nucleon** as the **average energy per nucleon needed to separate a nucleus into its individual nucleons.**
- If the **Binding energy per nucleon** is relatively large, the **nucleus is relatively stable.**

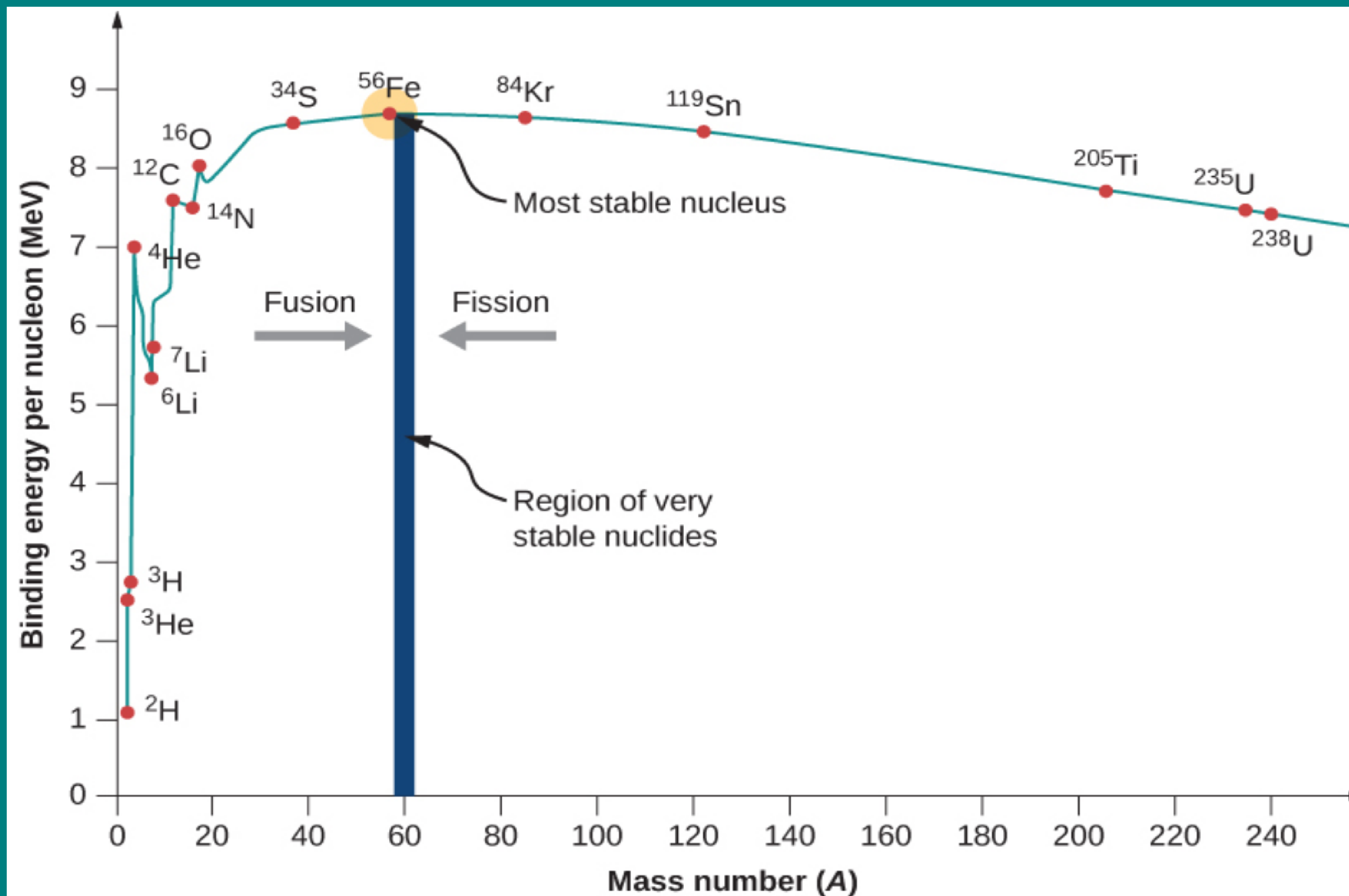
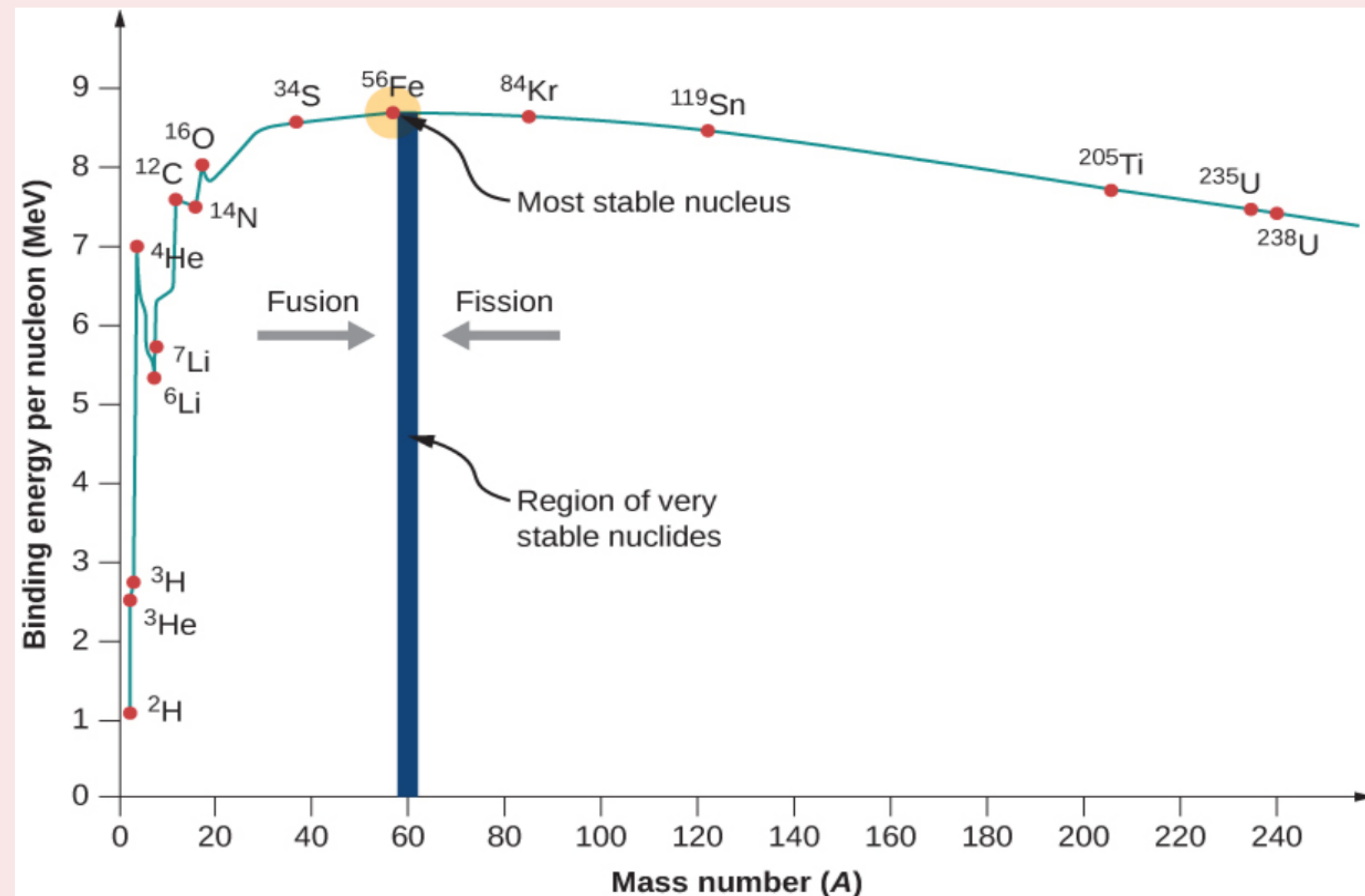


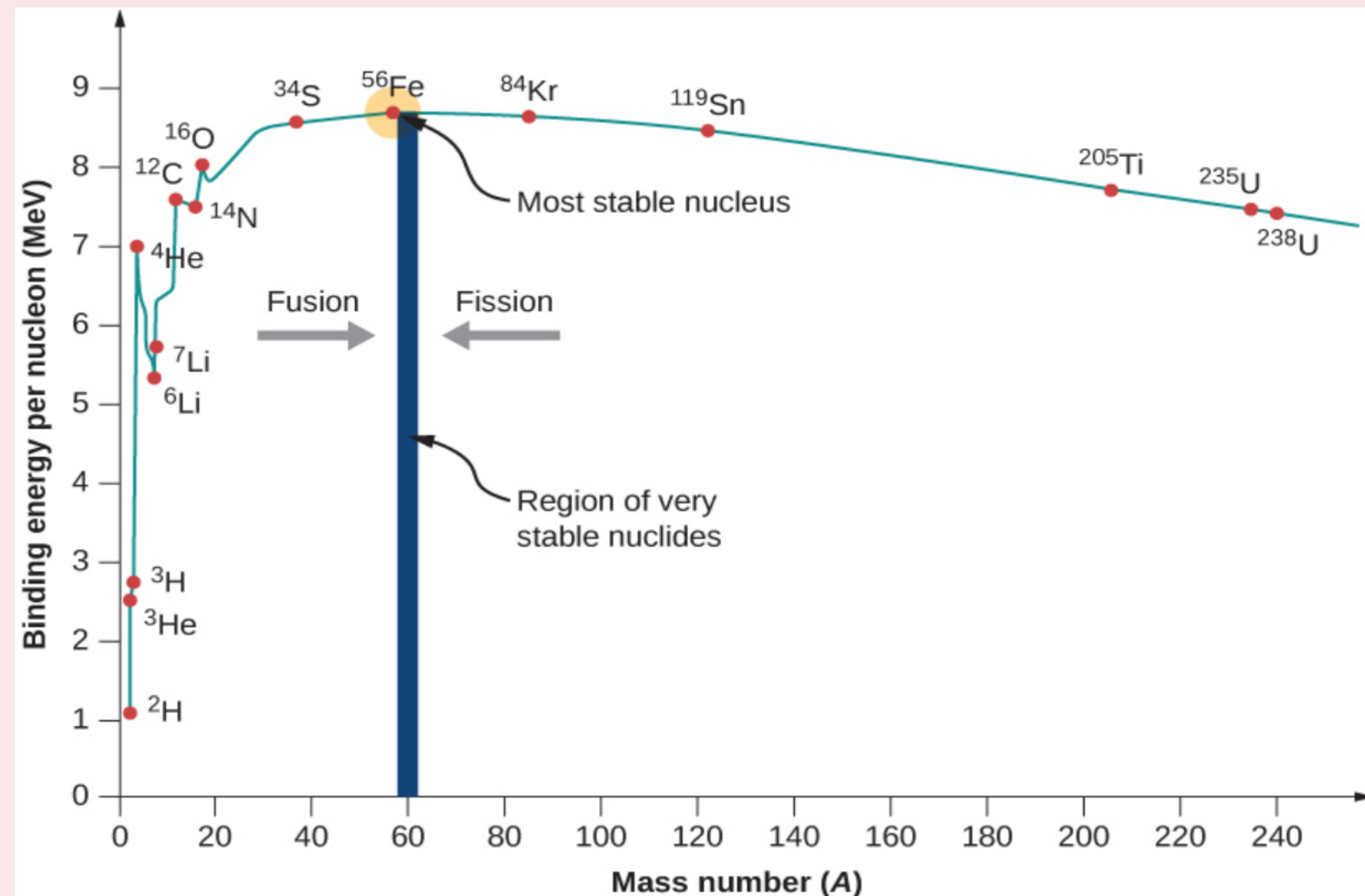
Figure 10.7 In this graph of binding energy per nucleon for stable nuclei, the BEN is greatest for nuclei with a mass near ^{56}Fe . Therefore, fusion of nuclei with mass numbers much less than that of Fe, and fission of nuclei with mass numbers greater than that of Fe, are exothermic processes.

Graph of Binding Energy per Nucleon

- We notice the following **main features of the plot**:

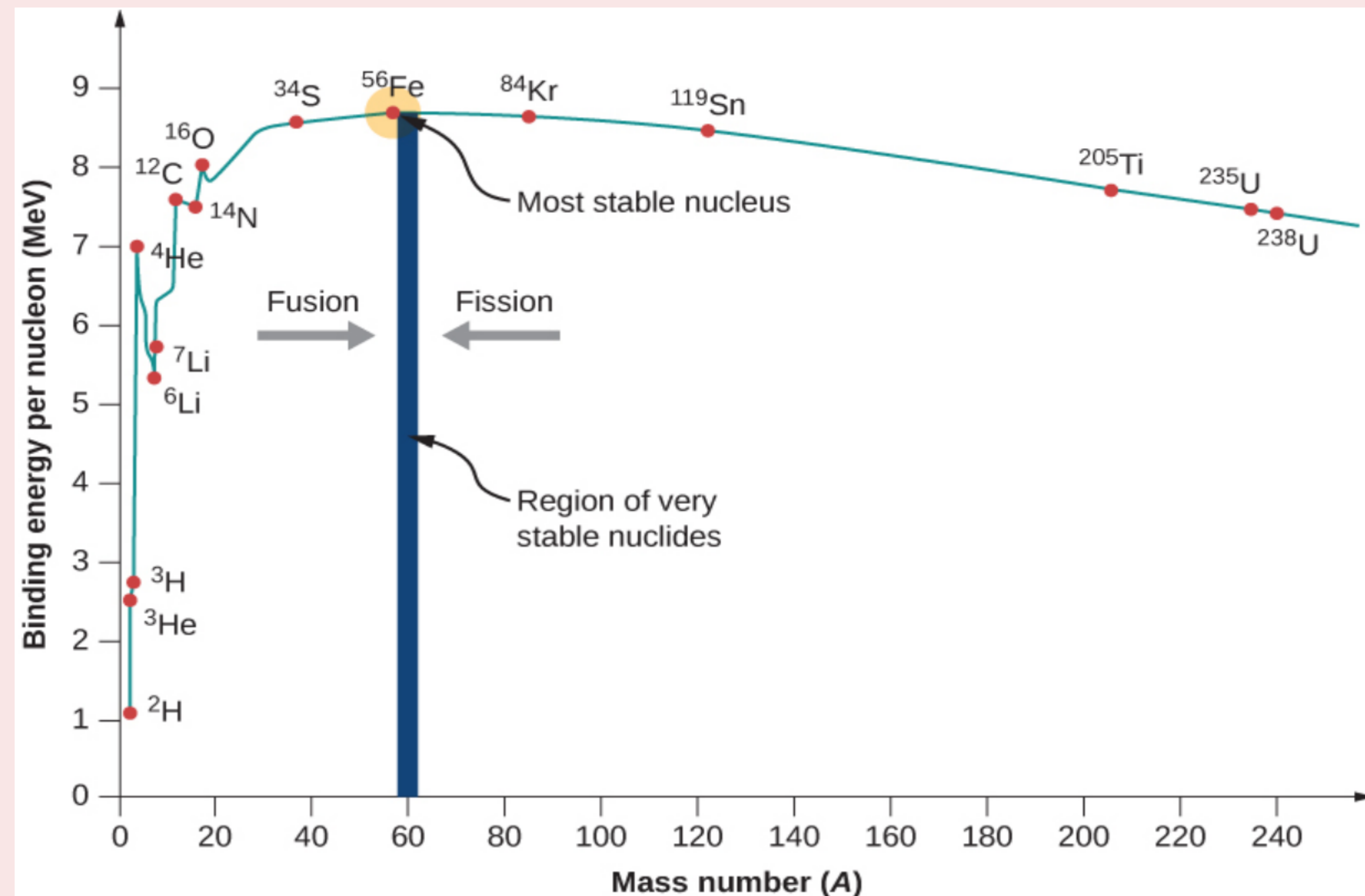


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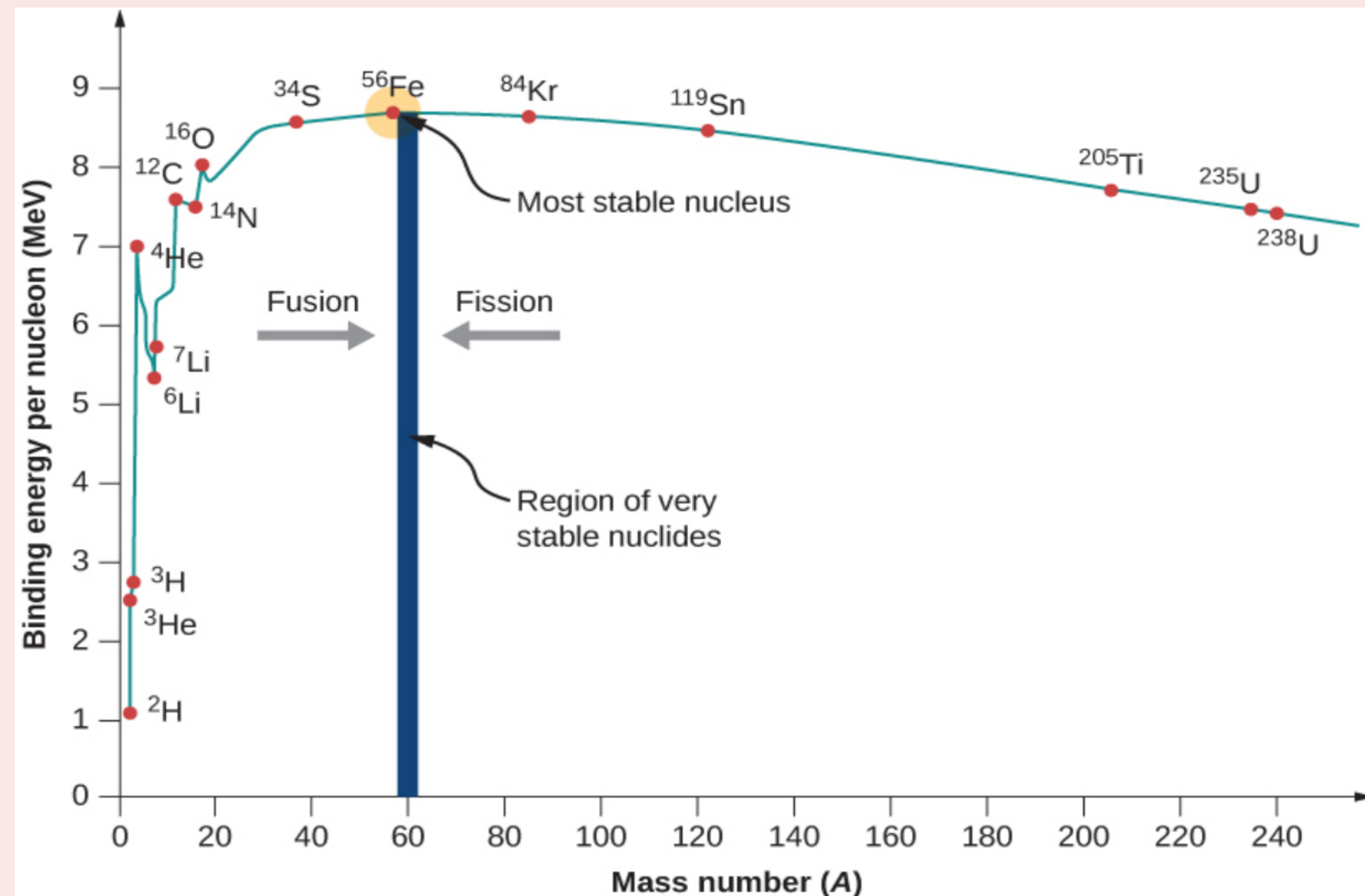
- We notice the following **main features of the plot**:
- the **binding energy per nucleon**, E_{bn} , is practically **constant**, i.e. practically independent of the atomic number of middle mass number nuclei ($30 < A < 170$).

Graph of Binding Energy per Nucleon



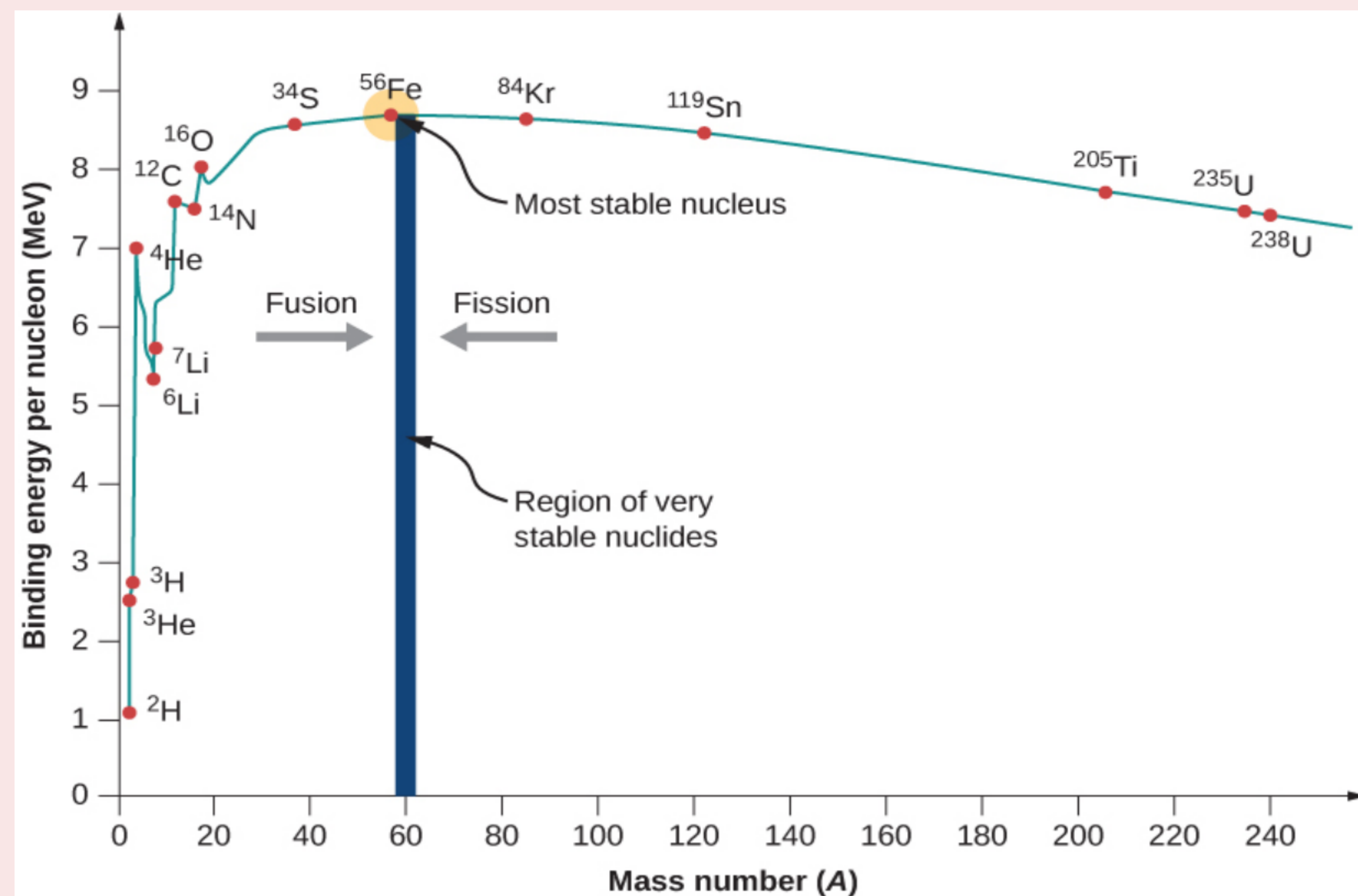
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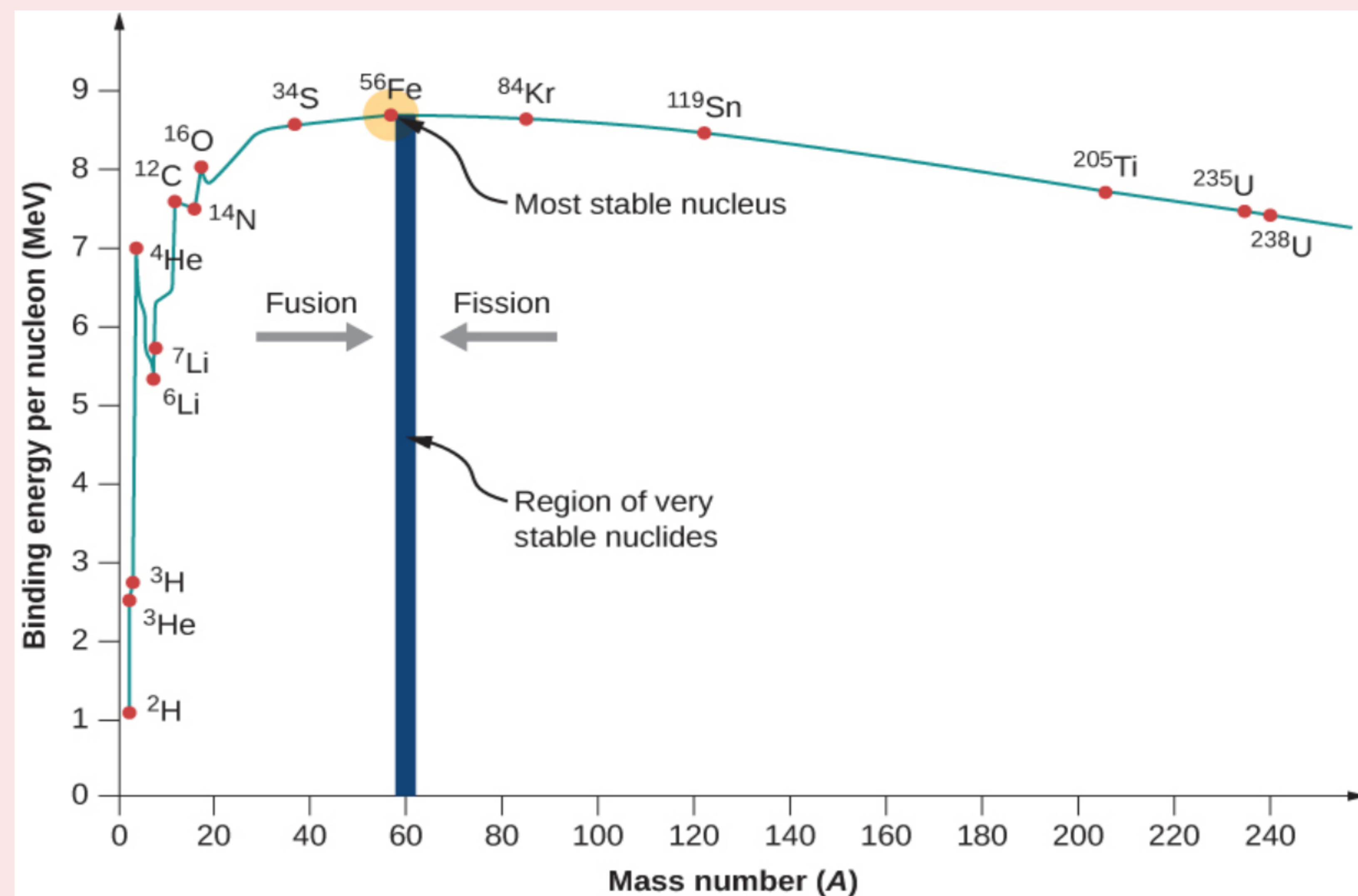
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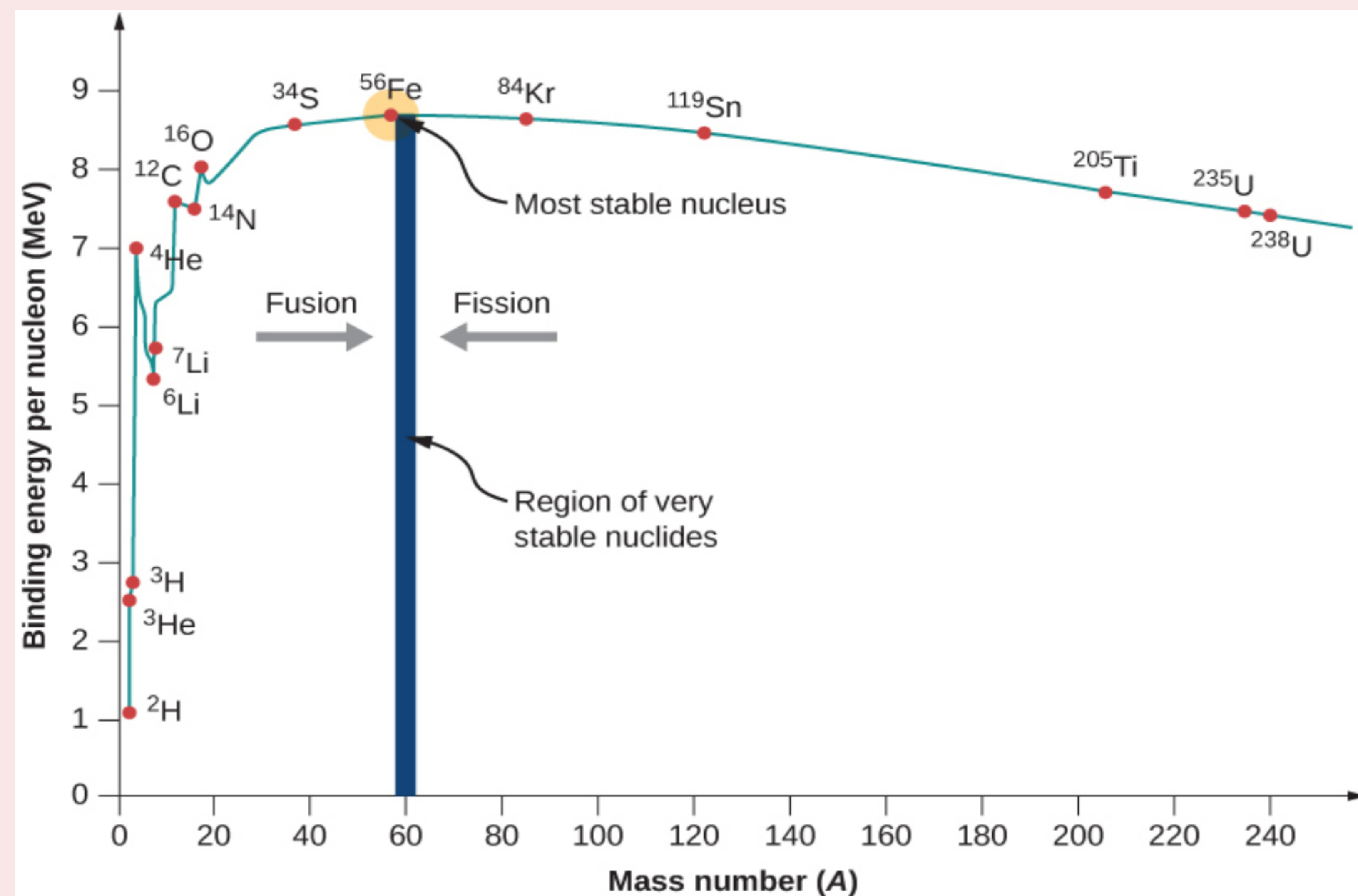
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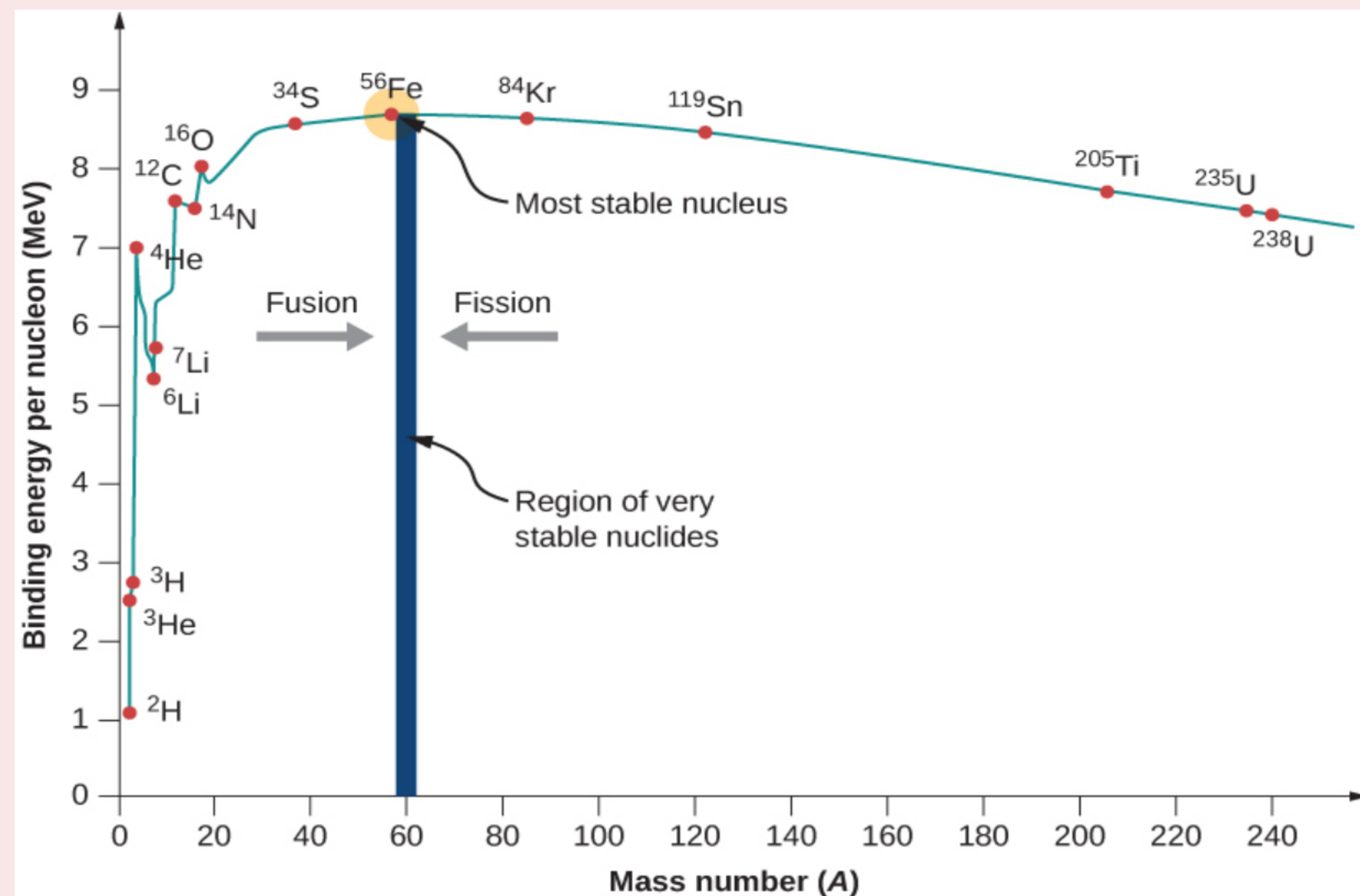
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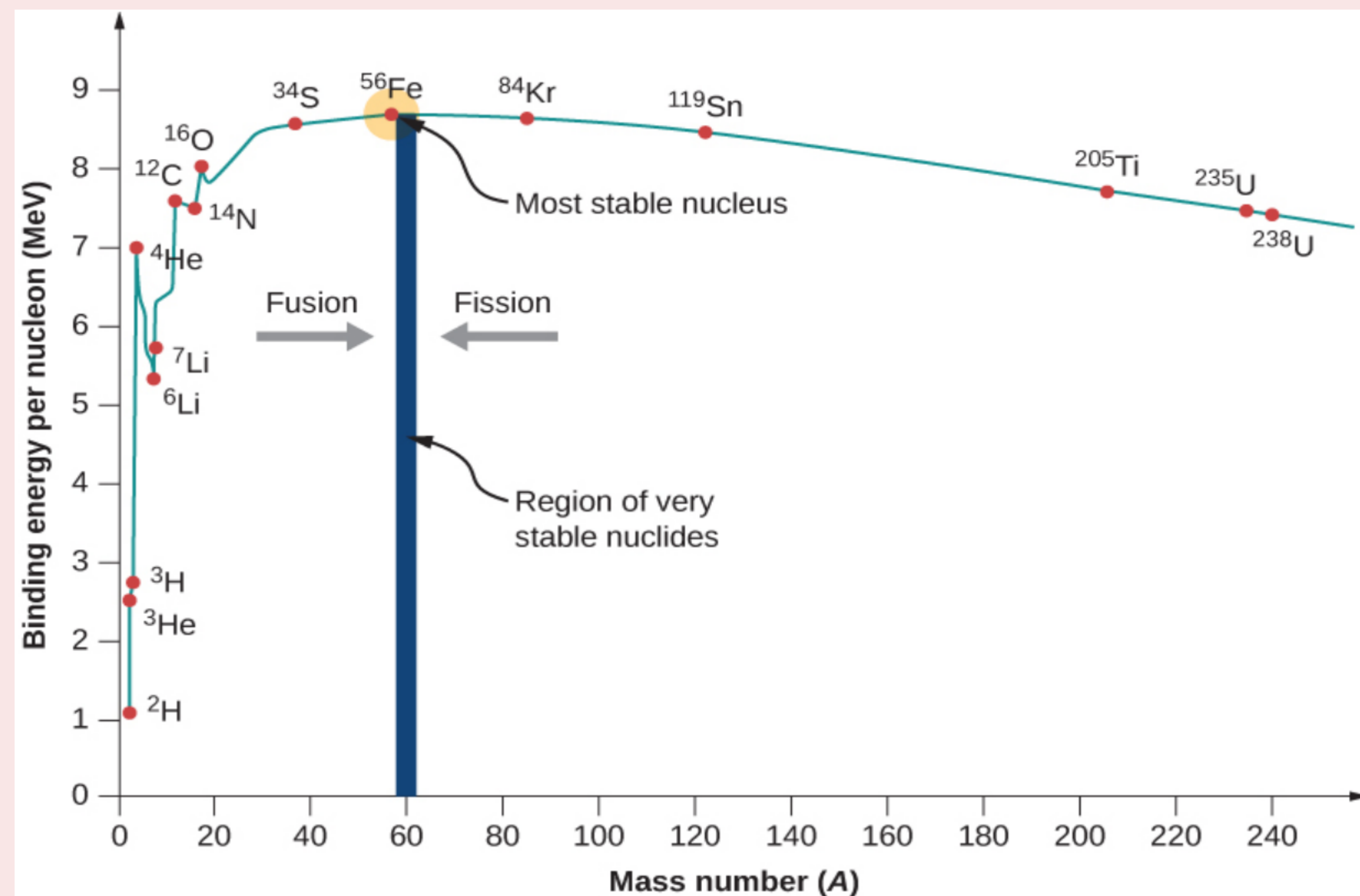
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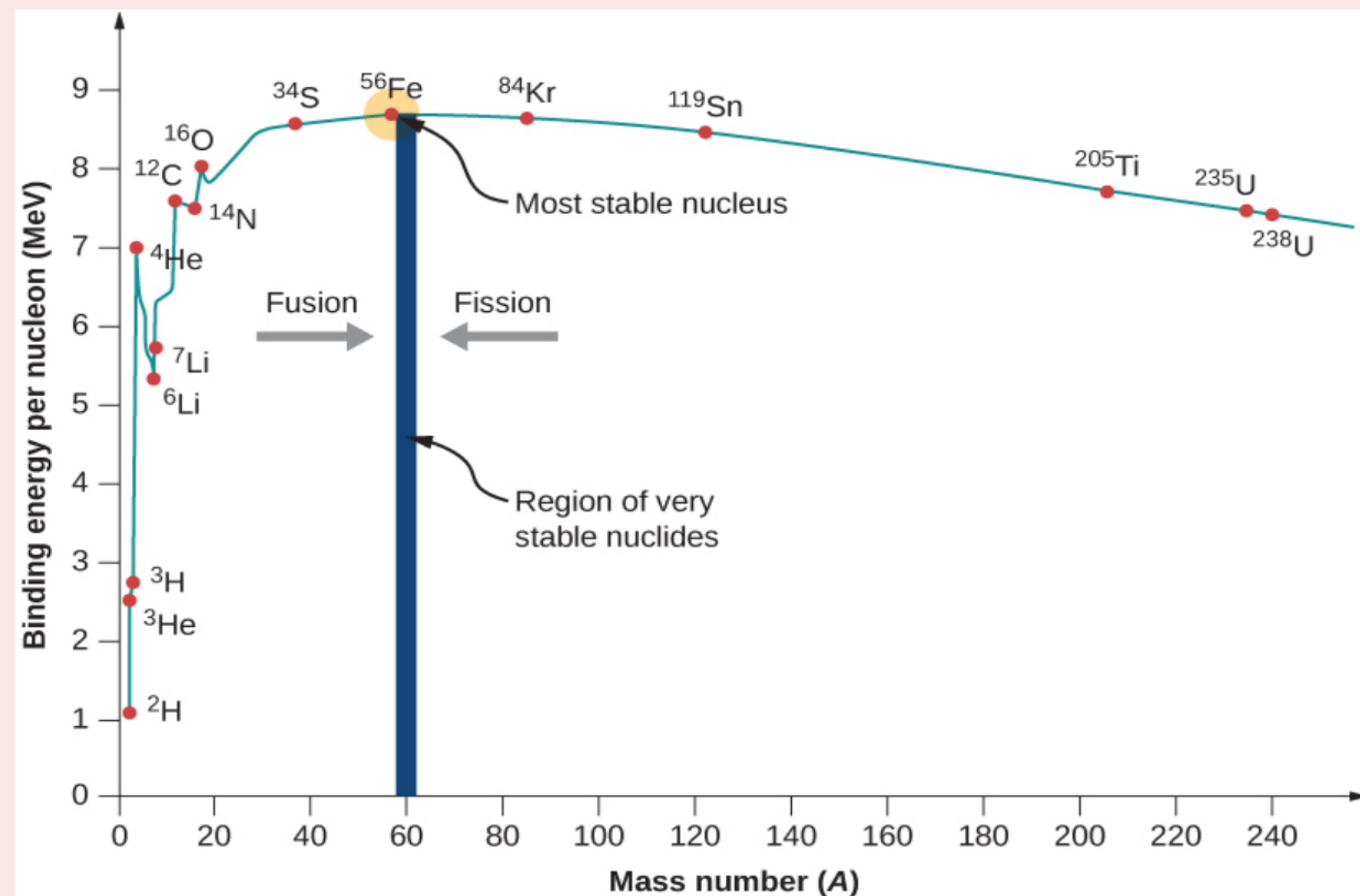
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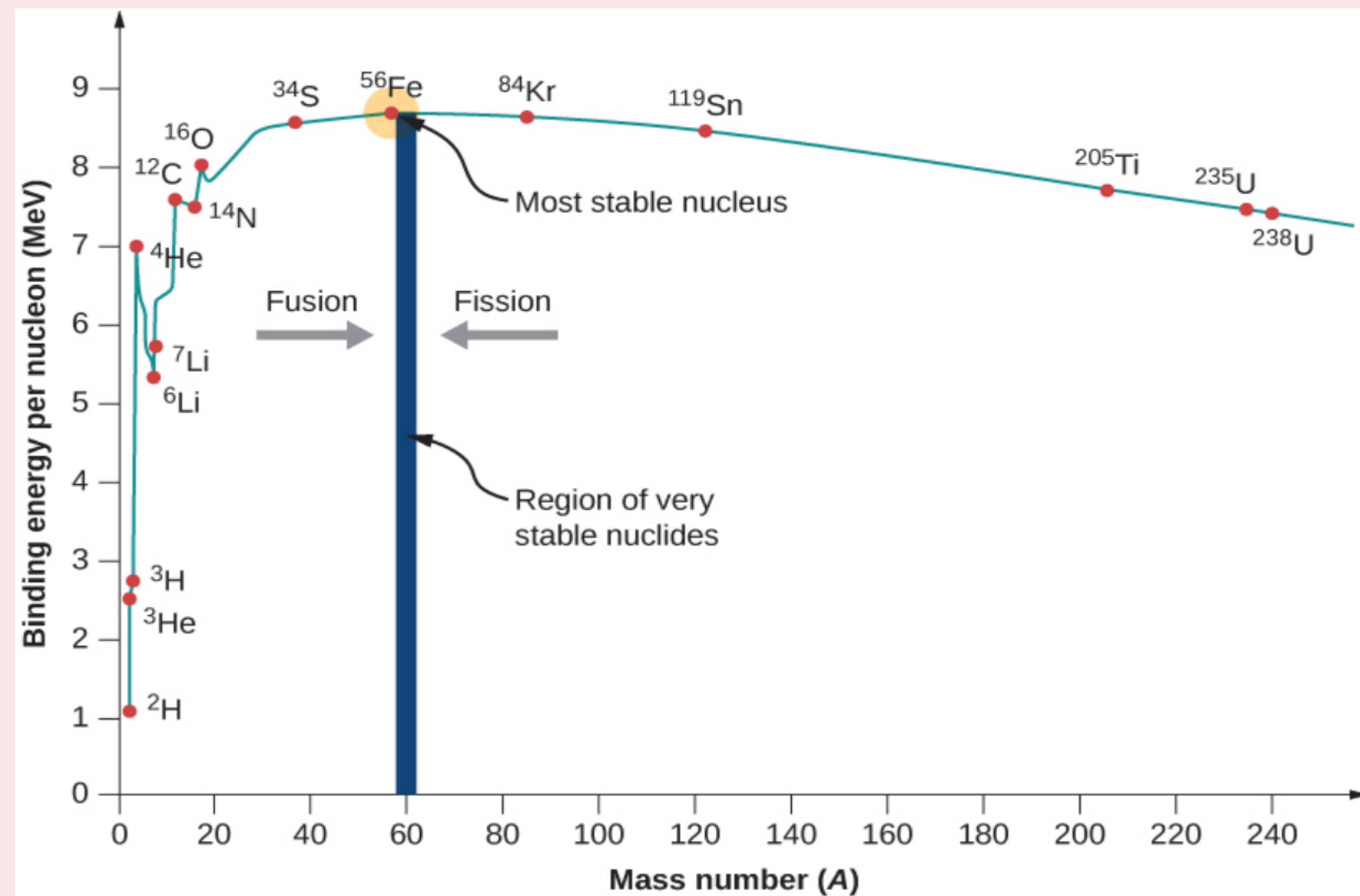
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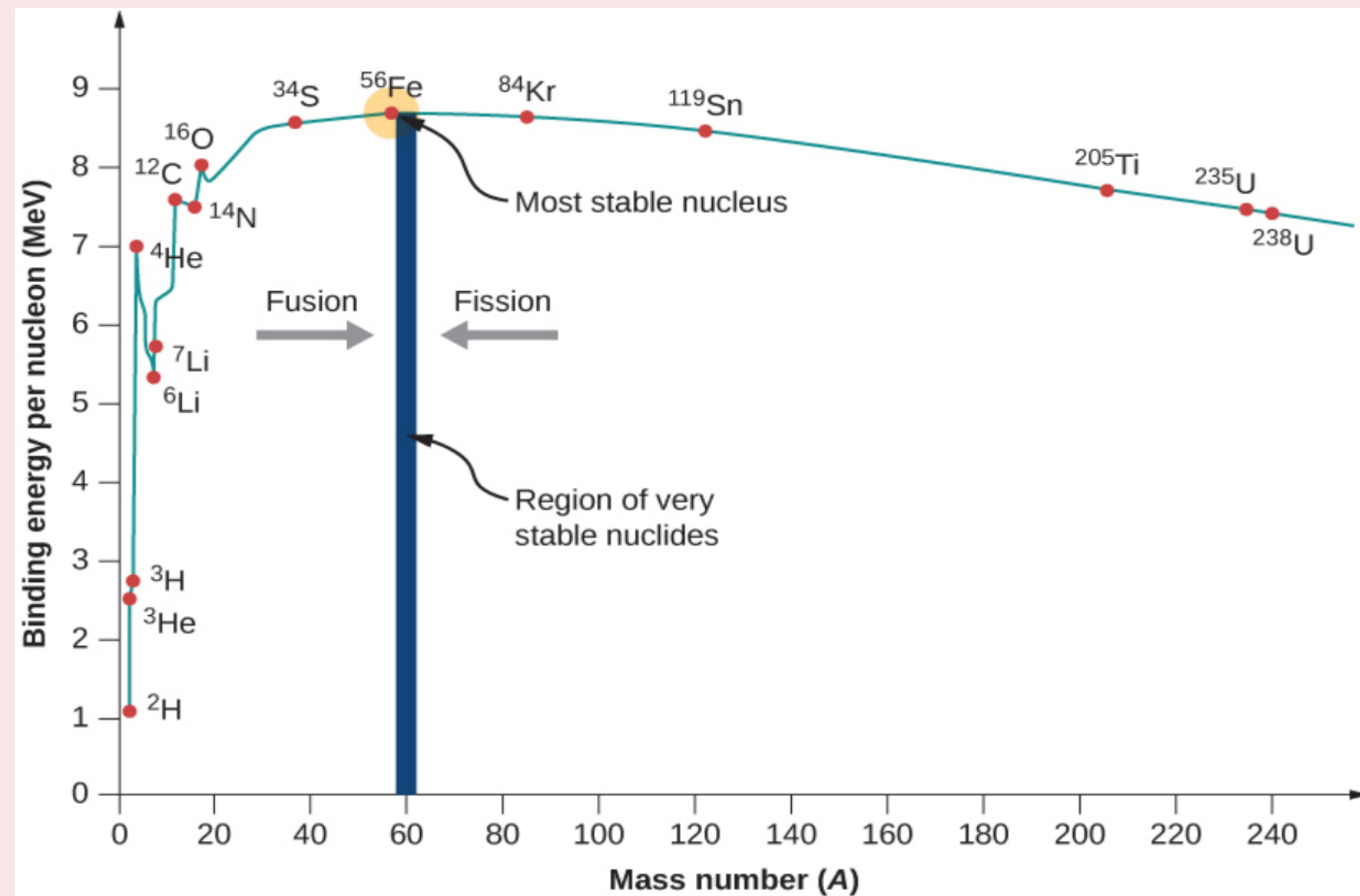
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- The reason the graph rises and tapers off has to do with **competing forces in the nucleus**.

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- But at **high values of A** , **repulsive electrostatic forces** between forces **begin to dominate**, and **these forces tend to break apart the nucleus** rather than hold it together.

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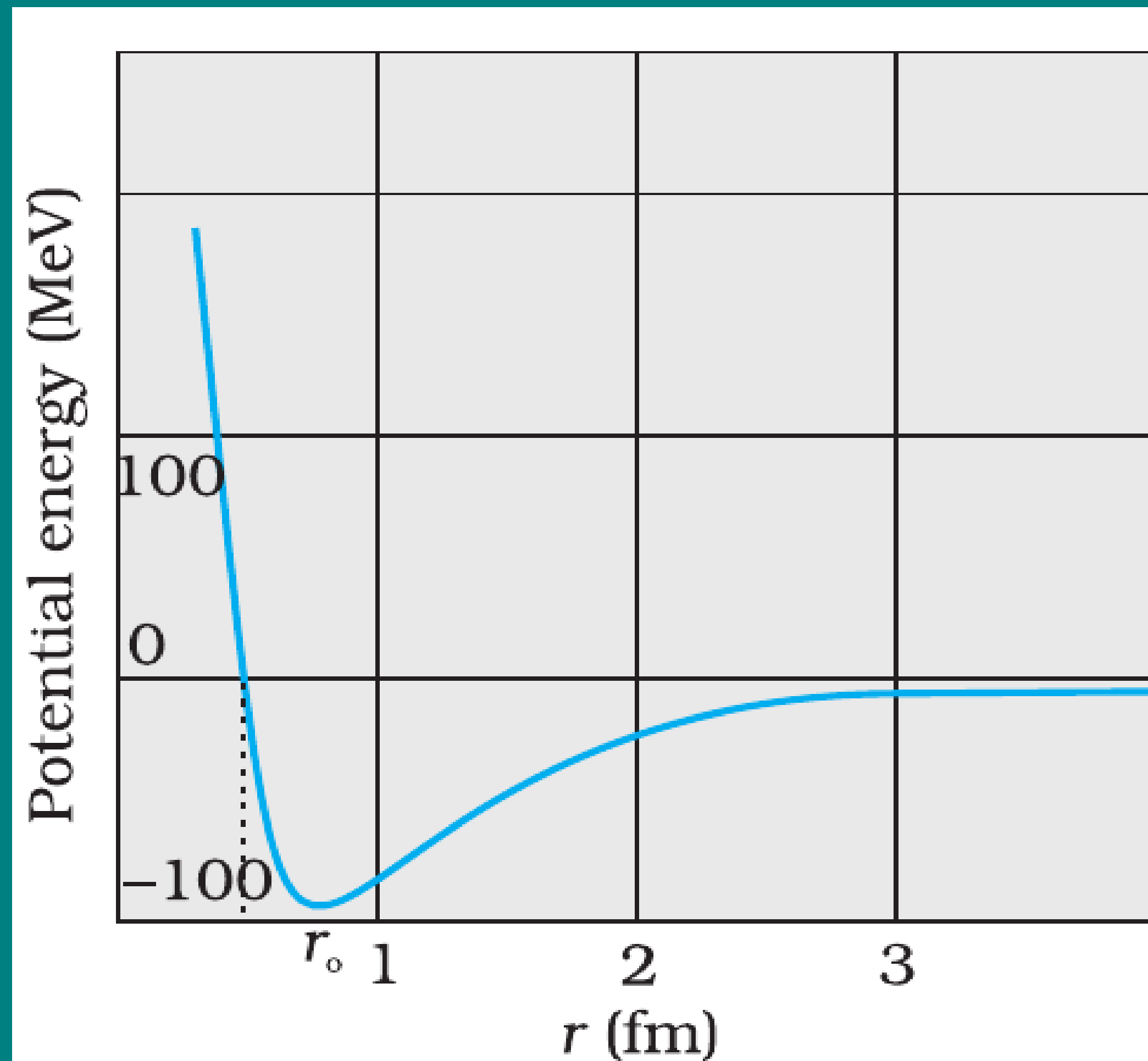


FIGURE 13.2 Potential energy of a pair of nucleons as a function of their separation.

For a separation greater than r_0 , the force is attractive and for separations less than r_0 , the force is strongly repulsive.

Nuclear Potential Energy vs Distance

- Potential energy minimum at ≈ 0.8 fm $\Rightarrow$ stable bound state.

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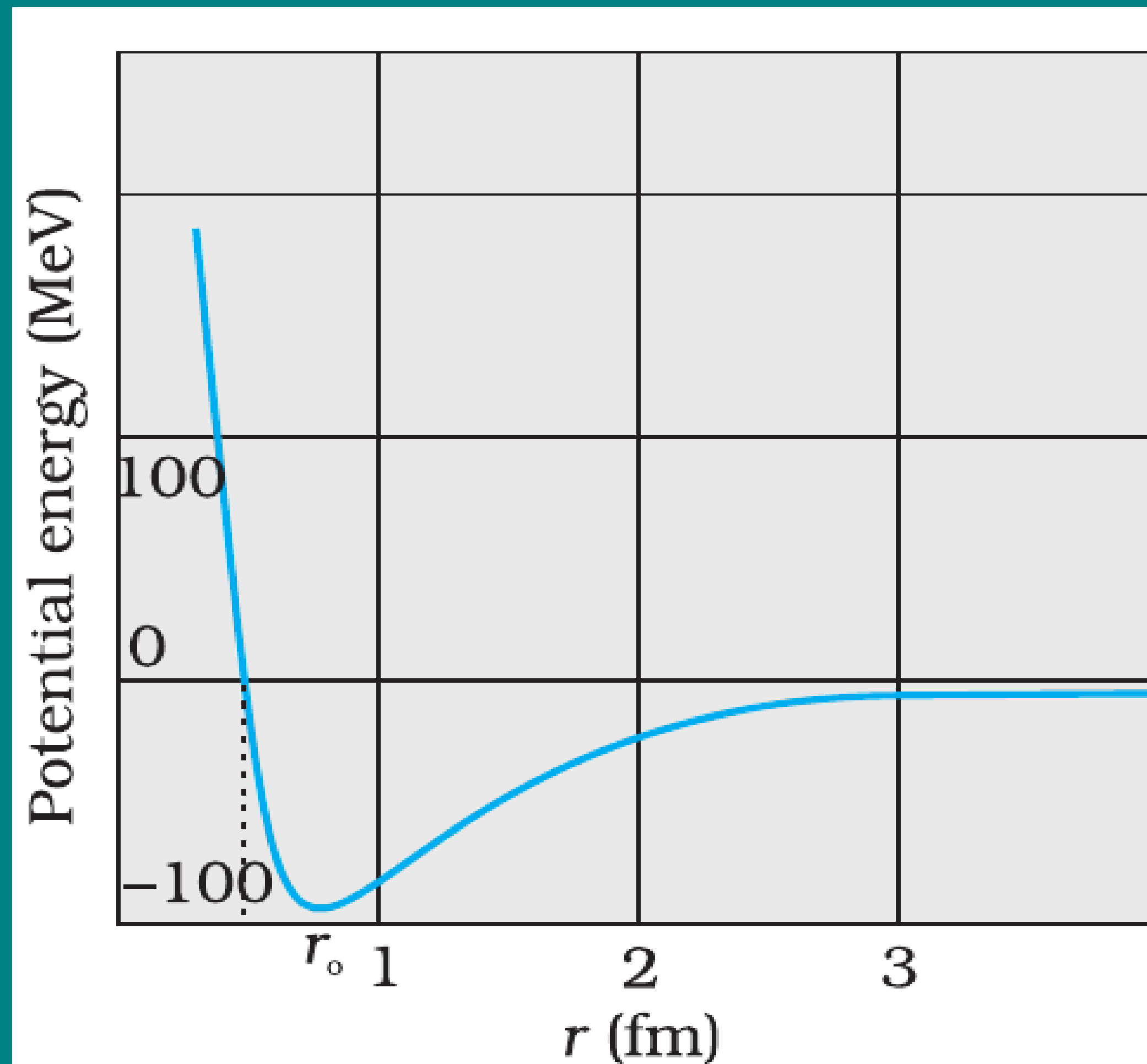


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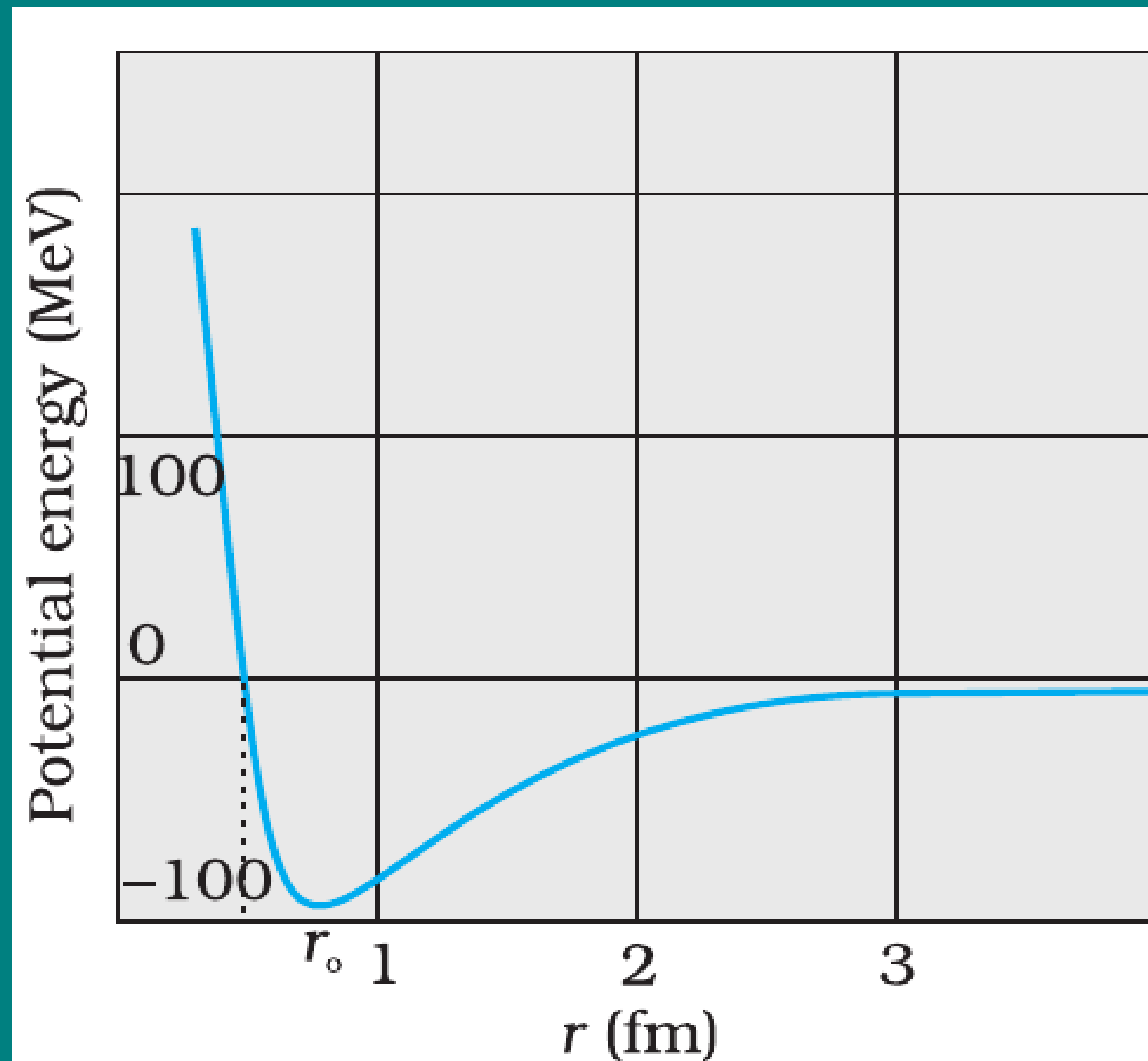


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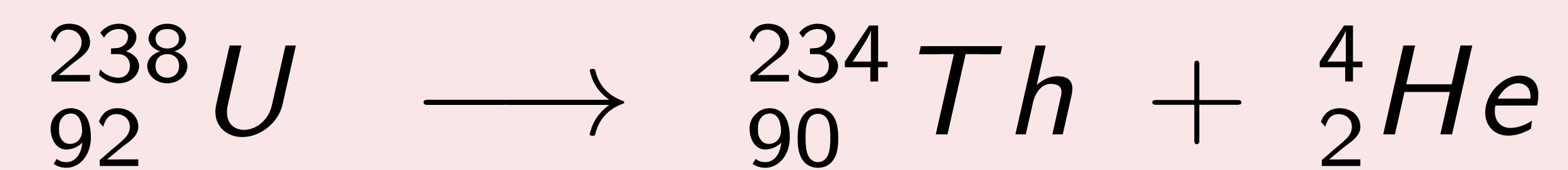
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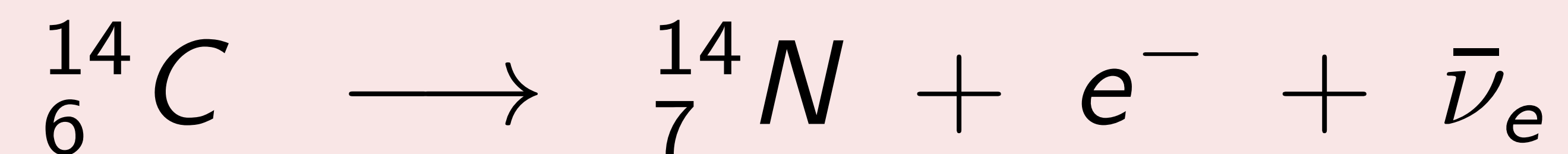
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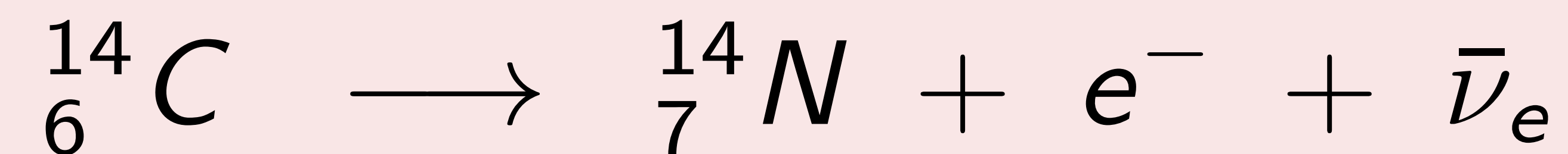
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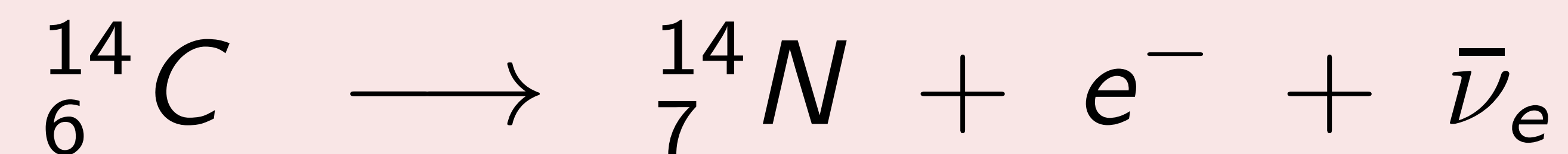


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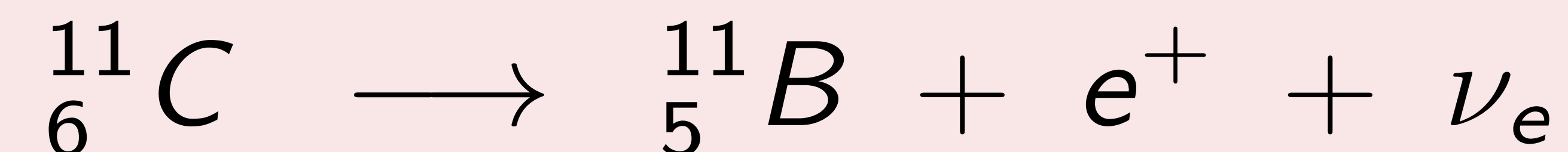
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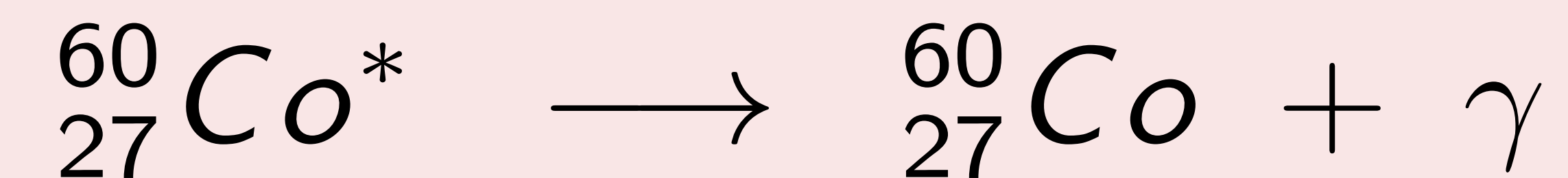
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(* denotes excited nucleus).

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 - Peaks at **Iron (Fe, A = 56)** → the **most stable nucleus**.
 - Lower for **very light (A < 30)** and **very heavy (A > 170)** nuclei.
- **Implication:**
 - **Heavy nuclei** release energy when split (**fission**).
 - **Light nuclei** release energy when combined (**fusion**).

Nuclear Fission

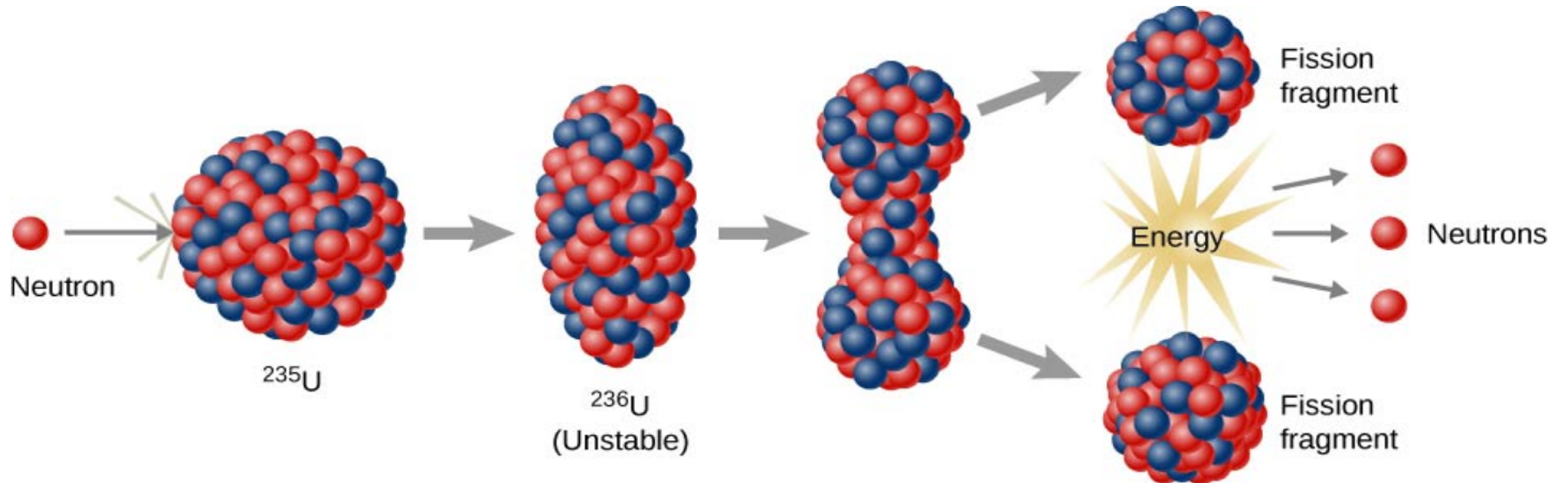


Figure 10.19 In the liquid drop model of nuclear fission, the uranium nucleus is split into two lighter nuclei by a high-energy neutron.

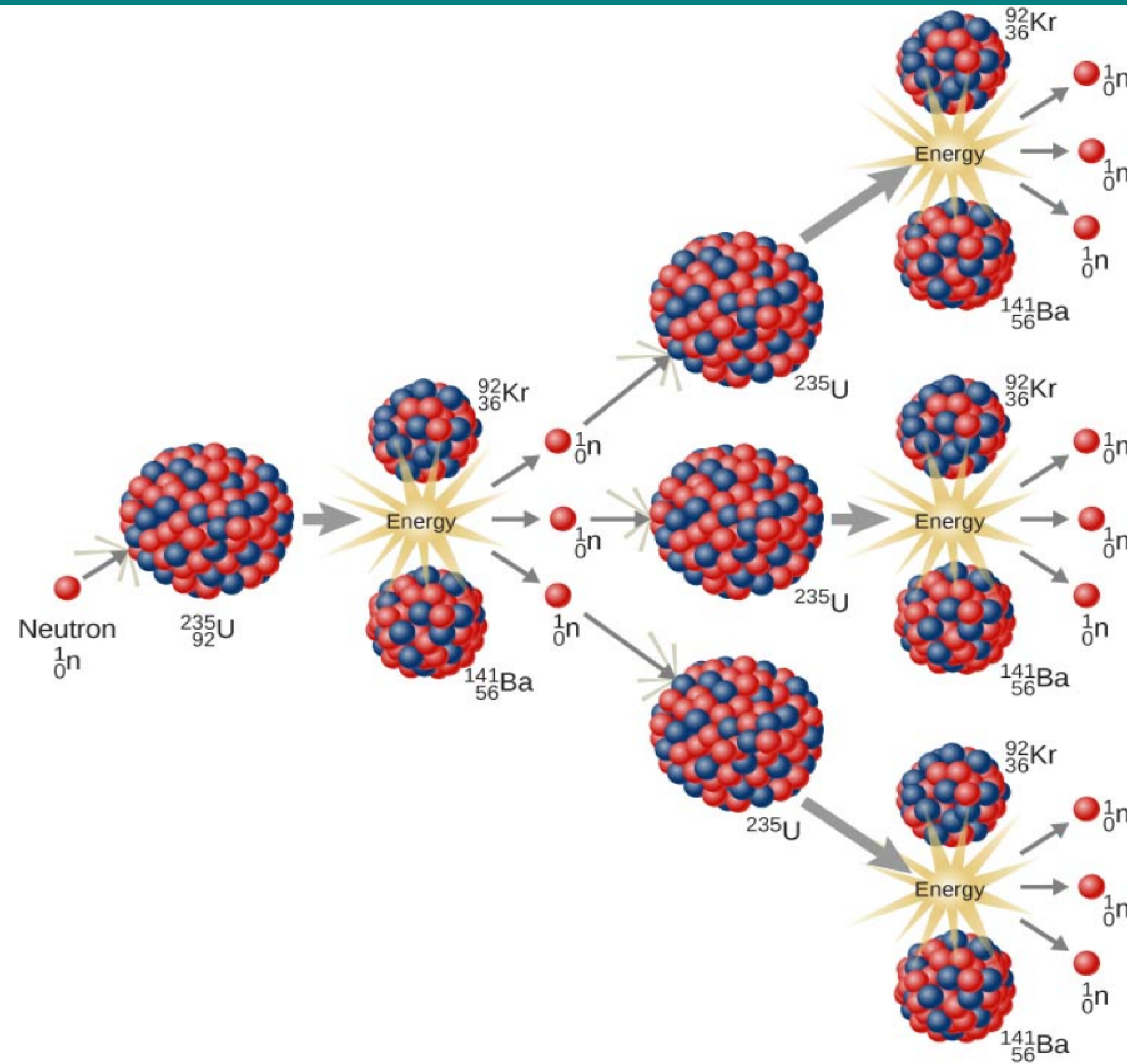


Figure 10.20 In a U-235 fission chain reaction, the fission of the uranium nucleus produces high-energy neutrons that go on to split more nuclei. The energy released in this process can be used to produce electricity.

Nuclear Fission

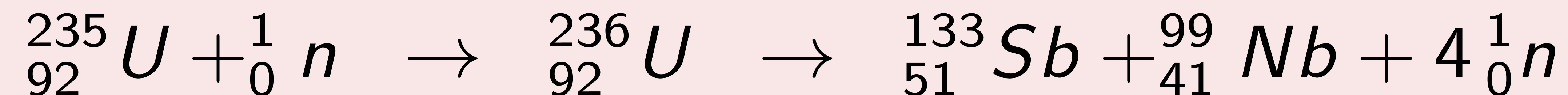
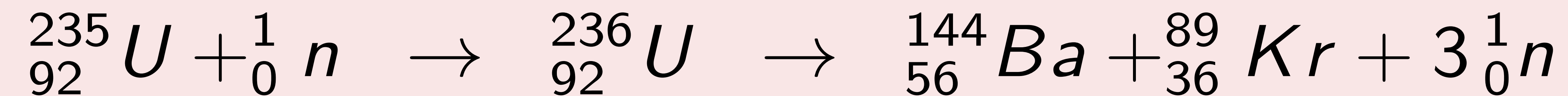
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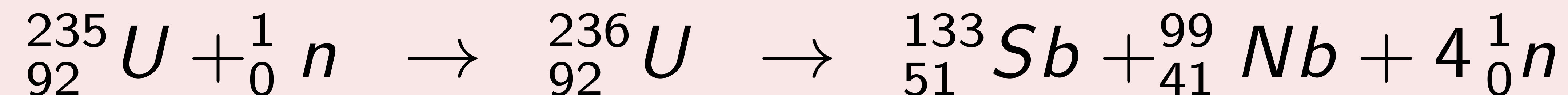
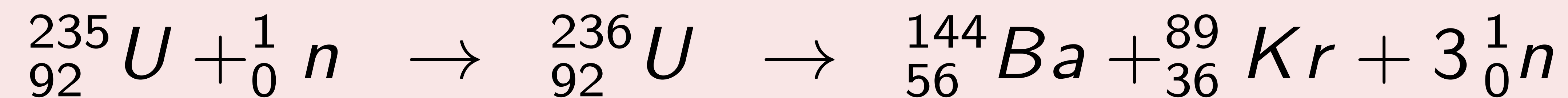
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- **Applications:**
 - **Controlled chain reaction:** Nuclear reactor (electricity).
 - **Uncontrolled chain reaction:** Atom bomb.

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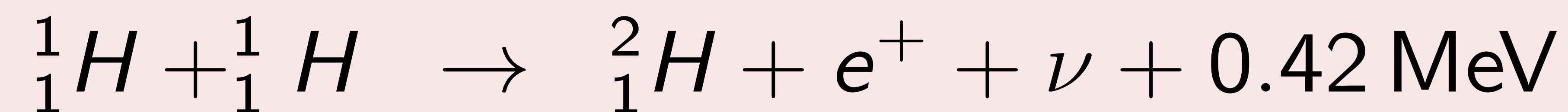
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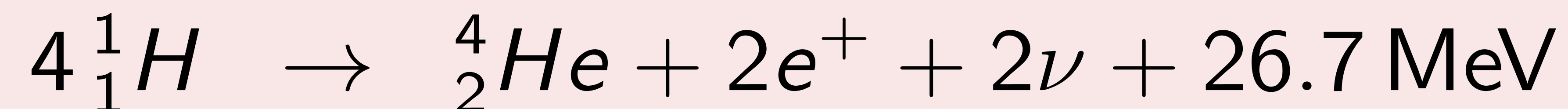
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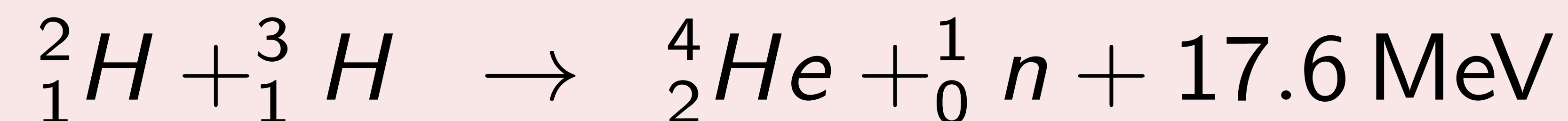
- **Proton–Proton Chain (in the Sun):**



Net reaction:



- **Deuterium–Tritium Fusion (on Earth, proposed):**



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